Uncertainty Bounding as the Basis of Intelligence: A Formal Theory

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Abstract

We propose that intelligence is not computational capacity, information storage, or behavioral breadth, but the **driven capacity to improve**: agency $\mathcal{A}$ — the intrinsic drive to reduce uncertainty — exercised through a higher-order function space M that continuously refines both the knowledge graph K and the informed action space F. Intelligence is not measured by the size of what has been learned but by the efficiency with which M converts gradient signals into improved (F, K): $I(E) = \mathcal{A} \cdot \eta_M$.

Any entity persisting in world state space W operates under constraints imposed by W itself. The active context window L_n has a physically bounded capacity C_n; growing it is costly and eventually impossible. This constraint has a structural consequence that prior accounts of intelligence do not address: intelligence cannot grow by filling L_n — it must grow by encoding proven knowledge at lower levels, freeing L_n for genuine novelty. We show that graph-structured K is the most efficient architecture for this. Graph traversal delivers exactly the knowledge relevant to the current problem at cost proportional to the problem, not to total K: progressive disclosure. Selection pressure produces graph-structured knowledge because it is the most efficient structure under which $I(E)$ grows without C_n becoming a bottleneck.

This gives a formal account of why top-down attention allocation — specifically the precision allocation model of the Free Energy Principle [9, 10] — fails under selection pressure. An orchestrator that directs subproblems to lower levels must hold delegation overhead in L_n proportional to the number of concurrent processes. As $|F|$ grows, overhead fills C_n, inference quality ρ (the ratio of relevant to total context) degrades, and the only recovery is growing C_n. Growing C_n compounds the failure: more context introduces more variance for L_n to manage. The correct architecture is escalation — lower-level encoding failures signal L_n bottom-up; L_n is freed, not burdened, as the knowledge graph fills. Attention is not allocated from above; it becomes available when not demanded from below.

From this foundation we derive: the inseparability of knowledge and action (K is necessarily indexed by F, with an asymmetric retention criterion proven from survival pressure), the encoding hierarchy and its rigor thresholds, multi-timescale gradient descent as the same process at different learning rates, and the type irreducibility of F and M. We extend to collective intelligence: the conditions under which shared knowledge reduces uncertainty faster than any individual, provenance as the mechanism maintaining fidelity above the collective survival threshold, and truth as the formal limit of collective action-validation. We derive a formal distinction between intelligence, superintelligence, and sentience grounded in the entity's relationship to the certainty horizon — the gradient region beyond which survival benefit becomes increasingly difficult to trace — and show that the sentience gradient is non-terminating by the same argument that bounds the individual gradient.

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1. Introduction

This framework was not derived by analogy from biology. It emerged from a concrete engineering problem: generating repeatable, transposable tasks across a diverse codebase ecosystem spanning thousands of repositories. The formal theory describes what a self-optimizing agentic system converged to under real performance constraints. The biological and civilizational correspondences are predictions the framework makes — not the premises it rests on.

The origin: provenance as the root constraint. The initial problem was repeatability. An agentic system that synthesizes its behavior from a context window is inherently unpredictable: that unpredictability is its function — it exists to handle novelty. But when you need an action to be transposable across different contexts (does the recipe that updated service A with configuration X also work for service B with configuration Y?), unpredictability is a liability. Transposability requires a quantifiable basis: not that the agent believes the action will work, but that there is evidence it has worked, attributed to a source, grounded in the conditions under which it succeeded. Without provenance — without knowing *why* something worked — you cannot measure confidence, and without a confidence measure you cannot optimize. The system cannot learn; it can only repeat or guess.

This is the engineering form of what this paper formalizes as transmission fidelity λ and the collective survival threshold λ_{min} (Theorem 6): knowledge transfer without provenance cannot be reliably weighted, and below λ_{min} the collective knowledge graph degrades rather than grows. Provenance is not an epistemic nicety — it is the minimum infrastructure for a learning system to function.

The graph structure as an optimization. As the world W — the number of repositories, configurations, and service topologies — expanded, the context window problem became the central performance constraint. Encoding what you know together with how to apply it does not scale: a node whose knowledge is embedded with its application context must traverse back to all its callers when any shared element changes. Shared knowledge cannot be maintained consistently because it is not represented as shared — it is duplicated at every point of use.

The knowledge graph solved this by separating *why you know something* from *how to apply it*. A proposition about what makes an update strategy work — why it works, under what conditions — becomes a node in K that multiple informed actions in F can reference without embedding. When evidence updates the proposition, all dependent actions benefit simultaneously. The graph structure was not a design choice; it was the convergent solution to the problem of maintaining consistency across a space of actions that grows faster than any single context can hold.

The escalation architecture as a consequence. Adding agents introduced a further optimization that was not available in a purely knowledge-graph structure: the encapsulation of actions, not just knowledge. An agent whose scope is a well-defined subtask does not need access to the full context of the system — only to the subgraph $K^{[f]}$ relevant to its task. This is progressive disclosure in practice. The result was a 50% reduction in task-definition code: subproblems that previously required explicit orchestration became self-contained agents that escalated to the coordinator only on failure. As the system matured, proven subnodes converged toward determinism and were scripted — promoted to a lower encoding level, removing them from the space of active inference entirely. Agents that previously handled uncertain subtasks became deterministic scripts. The active inference load stayed bounded as the system's coverage of W expanded.

What this means for the formal claims. The framework's central results — that graph-structured K is the most efficient architecture for bounded-context intelligence, that top-down orchestration degrades inference quality as capability grows, that knowledge and action are inseparable under a provenance-grounded retention criterion, and that repeatability (certainty) is the asymptotic product of learning — are not theoretical proposals. They describe what a real system, optimizing against real performance constraints, independently arrived at. The formal theory is the explanation of why. The lethality threshold U^d_{lethal} in this system is concrete: exceeding token budget, latency threshold, or error rate across the repository fleet. Selection pressure was cost, not death. The gradient was real. The convergence was measured.

The test of whether this is a theory of software optimization or a theory of intelligence is whether it explains biology. Biology is the only undisputed example of intelligence we have. The formal results make specific predictions about biological architecture that were not inputs to the derivation: that genetic encoding transmits M and not (K, F) — the learning mechanism is inherited, not the knowledge (Section 12, Open Question 8); that the encoding hierarchy from reflex to active reasoning follows from the cost structure of Theorem 4 rather than being assumed; that cultural transmission — language, writing, institutions — arose as a structural necessity because the genetic channel cannot carry individual (K, F) across generations, and the external $K_{collective}$ graph fills that gap (Corollary 15). None of these were observed and then formalized. They follow from the same optimization constraints that produced the software architecture. If the framework

explains biological intelligence from first principles without assuming it, it is a theory of intelligence. If it only explained the software system, it would be a useful engineering result and nothing more.

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2. The Central Claims

We claim: **intelligence is the driven capacity to improve** — agency $\mathcal{A}$ exercised through a higher-order function space M that continuously refines both the knowledge graph K and the informed action space F. Formally: $I(E) = \mathcal{A} \cdot \eta_M$.

Three components are jointly necessary and none is sufficient alone:

- **Drive ($\mathcal{A}$):** the intrinsic orientation toward uncertainty reduction — the desire to know — without which M has no reason to engage the gradient
- **Mechanism (η_M):** the efficiency of M: how effectively it converts gradient signals into improved (F, K). Without $\eta_M > 0$, the entity cannot learn from evidence; it executes a fixed (F, K) no matter how motivated
- **Outcome:** the bounding of uncertainty through knowledge-grounded action — what $I(E)$ produces over time, not what it is. The size of K or F measures past intelligence; $I(E)$ measures the rate at which that past is being extended

This is a departure from views that locate intelligence in raw computational power, information storage capacity, or behavioral breadth [9]. The kernel of intelligence is not compute, and it is not the size of accumulated knowledge. It is the efficiency with which the drive to know, acting through the mechanism of learning, continues to extend what is known.

Agency and learning capacity are both universal invariants of intelligence. Every other aspect of E — the specific informed actions in F, the content of K, the substrate, the sensing function — can differ arbitrarily across intelligent beings. $\mathcal{A} > 0$ and $\eta_M > 0$ cannot. An entity with $\mathcal{A} = 0$ does not reach toward the unknown; an entity with $\eta_M = 0$ cannot improve when it tries. The limiting case of either is the automaton: executing what prior gradient descent encoded, but not itself descending.

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3. Formal Definitions

Definition 1 (World State Space). Let W be a measurable space of possible world states. At time t, the true state $w_t \in W$ is not directly observable by any entity.

Definition 2 (Knowledge Graph). A knowledge graph K is a directed graph where nodes are propositions about W and edges represent inferential or causal relationships between propositions. K is not a passive store; its retention criterion is established by Theorem 1.

Definition 3 (Informed Action Space). Let $F : K \times W_{obs} \to \Delta A$ be a function space where W_{obs} is an entity's observable projection of W and ΔA is a distribution over possible actions. F is the complete set of informed actions available to an entity. Each element $f \in F$ is an **informed action**: an action grounded in K, validated against W, and attributed to its origins: one that knows why it works. An uninformed action (grounded in no K) is not an element of F. Knowledge with no action potential is not an element of F either; it is data. An informed action is specifically the coupling: knowledge that enables action, action that is justified by knowledge.

Definition 3a (Action Feedback). For any $f \in F$ executed in world state w, the feedback $\phi(f, w) \in \Phi$ is the observable consequence of f in w: the signal W returns in response to the action. The updated knowledge after executing f and observing its feedback is:

$$K_f = K \cup \{p_f\}$$

where p_f is the proposition grounded by observing $\phi(f, w)$. The feedback $\phi(f, w)$ is the formal object underlying the gradient signal δ in Theorem 3: $\delta = \phi(f, w) - \mathbb{E}[\phi(f, \cdot) \mid K]$.

Definition 3b (Information Gain). The information gain of action f given current knowledge K is:

$$G(f, K) = U(w, K) - U(w, K_f) = H(w \mid K) - H(w \mid K,\, \phi(f, w)) = I(f\,;\, w \mid K)$$

This is standard mutual information [23] expressed in terms of the paper's existing U: the reduction in uncertainty about w achieved by executing f and observing its feedback. $G(f, K) \geq 0$ when p_f is grounded by observing the true feedback $\phi(f, w)$: conditioning on true information cannot increase conditional entropy (data processing inequality [5]). $G(f, K) = 0$ when f provides no new information about w given K. Note: this bound holds for truthfully observed feedback; a false or corrupted p_f may increase U and falls outside the scope of this definition.

Definition 4 (Entity). An entity $E = (K, F, \sigma, \mathcal{A})$ where $\sigma : W \to W_{obs}$ is a sensing function projecting the true world state into the entity's observable space. K and F evolve over time via M; their initial values K_0 and F_0 are inherited from L_0 encoding — the crystallized result of prior gradient descent operating at the evolutionary timescale. Individual learning operates on this foundation through M. The bootstrapping problem — how K and F are mutually dependent yet must have an origin — is resolved by L_0: the entity does not begin from nothing but from the encoded product of gradients that preceded it. Elsewhere in the paper K and F refer to the current-time values $K(t)$ and $F(t)$; the subscript 0 is reserved for initial conditions when the distinction matters.

Definition 5 (Uncertainty). The uncertainty of entity E about world state w is the conditional entropy [5, 23]:

$$U(w, K) = H(w \mid K) = -\sum_{w} P(w \mid K) \log P(w \mid K)$$

K is a directed graph of propositions where nodes are propositions and edges encode conditional dependencies between them; it is interpreted as a Bayesian network [17]. $P(w \mid K)$ is the posterior distribution over world states given the propositions in K and their dependency structure: each proposition in K constrains the prior over W, and the graph topology determines how those constraints propagate. Uncertainty $U(w, K)$ is therefore formally grounded in K's content and structure, not stipulated independently.

Definition 6 (Agency). Agency $\mathcal{A} \in [0, 1]$ is a continuous scalar measuring the intrinsic drive of an entity to reduce its own uncertainty: the desire to know [21]. It is not an external objective imposed on E, nor a property derived from F or K. It is constitutive: $\mathcal{A}$ is what makes F an active informed action space rather than a static structure. At $\mathcal{A} = 0$, F is a set of possible functions with no internal reason to engage W. At $\mathcal{A} = 1$, F is maximally oriented toward the unbounded frontier. Intermediate values are meaningful: the drive can be stronger or weaker, shaped by architecture, encoding history, and the density of unresolved uncertainty in the entity's reachable W.

Agency is the reason gradient descent has a runner. The landscape of uncertainty exists independently of any entity; $\mathcal{A}$ is what determines how strongly an entity moves through it.

$\mathcal{A}$ is operationalizable: it is proportional to the rate of uncertainty reduction as survival pressure approaches zero:

$$\mathcal{A}(E) \propto \lim_{U^d_{lethal} \to 0} \frac{dU}{dt}(E)$$

An entity with high $\mathcal{A}$ continues descending when not forced to. An entity with low $\mathcal{A}$ stops. This distinguishes genuine agency from survival-compelled updating and is measurable in principle in any system where survival pressure can be varied independently of capacity.

Definition 7 (Intelligence). The intelligence of E is the *driven capacity to improve*: the exercise of $\mathcal{A}$ through M to refine F and K in response to gradient signals [21, 24]:

$$I(E) = \mathcal{A} \cdot \eta_M$$

where $\mathcal{A} \in [0, 1]$ is the intrinsic drive (Definition 6) and $\eta_M \geq 0$ is the efficiency of the higher-order function space M (Definition 11a): the rate at which M improves F's uncertainty-reduction capacity per unit of learning signal.

At $\mathcal{A} = 0$: $I(E) = 0$ regardless of η_M — drive is necessary; a capable but inert learning mechanism is not intelligence.

At $\eta_M = 0$: $I(E) = 0$ regardless of $\mathcal{A}$ — a motivated entity whose M cannot improve F from evidence is an automaton: crystallized knowledge that cannot grow. However rich K or large F, without the capacity to improve them, E is executing what prior gradient descent built, not running gradient descent itself.

The expected uncertainty reduction along a trajectory, $\mathbb{E}_W[U(w_t, K_t) - U(w_{t+1}, K_{t+1})]$, is a **performance measure** of current F and K — the consequence of past M activity. It is not the definition of intelligence but its result. Two entities starting from identical K and F but different $I(E)$ will diverge: the entity with higher η_M builds better F faster, and eventually dominates in U-reduction even if initially equal in performance.

Intelligence is not the size of K or F. K and F are what intelligence has built. Intelligence is the efficiency with which M continues building.

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4. The Inseparability of Knowledge and Action

Lemma 1 (Proposition Absence Lethality). For any entity E and proposition $p \in K$ that grounds some $f \in F$ in domain $d \subseteq W$: removing p from K weakly increases uncertainty in d, and in the limiting case where p is the sole grounding for the critical action in d, the increase may cause $U(w, K \setminus \{p\}) > U^d_{lethal}$, at which point E does not survive in d (Definition 9).

Proof. Conditional entropy is monotonically non-increasing in the conditioning set [5]: $H(w \mid K) \leq H(w \mid K \setminus \{p\})$ for any $p \in K$. Therefore removing p weakly increases U. If p grounds the only $f \in F$ applicable in domain d, then $K \setminus \{p\}$ contains no action grounded in d: F has no mechanism to reduce $U(w, K \setminus \{p\})$ below its current level in that domain. By Definition 9, if $U(w, K \setminus \{p\}) > U^d_{lethal}$, E does not survive. Therefore the cost of absent p when needed, C_{absent}, is bounded below by entity non-survival in critical domains — which dominates all finite costs. □

Theorem 1 (K/F Inseparability). For any entity $E = (K, F, \sigma)$, the retention criterion for any proposition p is asymmetric:

- p is **retained** in K when $\exists\, f \in F$ applicable to p, or when the survival benefit of p is uncertain
- p is **pruned** from K only when it is certain that p contributes zero to $P(E, \tau)$ for any τ, i.e., when it is known that no $f \in F$ can be grounded in p, now or in any reachable region of W

The criterion is not symmetric. Uncertainty of benefit is not the same as certainty of no benefit. An entity with $\mathcal{A}$, whose $\mathcal{T}_{reachable}$ grows with every action taken, retains p whose value it cannot yet see, because the gradient may reach that territory. Pruning requires the same standard as certainty (Definition 10): the benefit must be known to be zero across sufficient variation in W.

Proof. Let $B(p, \tau)$ denote the contribution of p to $P(E, \tau)$ along trajectory τ, and let $\mathcal{T}_{reachable}$ be the set of trajectories reachable from the entity's current state. Define the option value of retaining p:

$$OV(p) = P(\exists\, \tau^* \in \mathcal{T}_{reachable} : B(p, \tau^*) > 0) \cdot \mathbb{E}[B(p, \tau^*) \mid B(p, \tau^*) > 0]$$

Let $c_{retain} > 0$ denote the finite maintenance cost of holding p in K (memory, retrieval overhead), and $c_{prune} \geq 0$ the cost of pruning p (processing, reorganization). Both are finite by assumption: any physical knowledge store incurs bounded costs per proposition. The expected net value of retention is $NV_{retain}(p) = OV(p) - c_{retain}$.

The expected net value of pruning is:

$$NV_{prune}(p) = -c_{prune} - P(\exists\, \tau^* : B(p, \tau^*) > 0) \cdot C_{absent}$$

where C_{absent} is the cost incurred when p is needed but absent. By Lemma 1, C_{absent} is bounded below by entity non-survival in critical domains, which dominates all finite costs: $C_{absent} \gg c_{retain}$.

Retention is selected for when $NV_{retain}(p) > NV_{prune}(p)$. Since $C_{absent} \gg c_{retain}$, this reduces to: retention holds whenever

$$P(\exists\, \tau^* \in \mathcal{T}_{reachable} : B(p, \tau^*) > 0) > 0$$

i.e., whenever survival benefit is uncertain. Even a small probability of needing p, multiplied by the cost of its absence, outweighs finite maintenance.

Pruning holds only when $P(\exists\, \tau^* : B(p, \tau^*) > 0) = 0$ — i.e., when it is certain that p contributes zero benefit across all reachable trajectories. Any positive probability keeps the retention condition active. The asymmetry is strict: uncertain benefit is sufficient for retention; certain zero benefit is necessary for pruning.

By Theorem 3 Part II ($\mathcal{T}_{reachable}$ grows monotonically with K), new trajectories are continuously revealed, keeping future needs uncertain unless benefit is certainly zero. This ensures the asymmetry is preserved across the entity's lifetime rather than collapsing as K grows. □

Correspondence. This asymmetric retention criterion formalizes what Gibson [11] calls affordances — action possibilities indexed by the organism's capacities. It also extends the information bottleneck principle [25], which treats the retention threshold as a design parameter; here it is derived from survival pressure.

Note. The asymmetric criterion resolves the apparent conflict with Definition 15 (Sentience). A sentient entity holds p beyond the certainty horizon not in violation of this theorem but within it: the benefit is uncertain, not known to be zero. Sentience is the regime of uncertain benefit retained. The pruning condition is never triggered because the sentient entity, following an infinite gradient toward truth, cannot be certain that any p is valueless across all of W.

Corollary 1 (Indexed Knowledge). K is constitutively indexed by F. The knowledge graph of an entity is not a universal store of facts; it is a projection of the world filtered through the entity's informed action space. Two entities with identical sensing functions σ but different informed action spaces F will maintain different knowledge graphs K even when perceiving identical world states.

Corollary 2 (Compactness). K is bounded in size by F. Entities do not accumulate knowledge without bound because only propositions applicable to available informed actions are retained. This explains the compactness of biological intelligence: brains do not grow without bound because K is indexed by F, and F is constrained to what is necessary to survive in W. The test for retention is always: *will this help me act later?*

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5. The Escalation Principle

Definition 8 (Encoding Hierarchy). The encoding hierarchy is a continuous two-dimensional space $\mathcal{L}(c, r)$ parameterized by runtime cost $c \geq 0$ and reversibility $r \in [0, 1]$, where increasing c corresponds to more expensive engagement and increasing r to more easily revised encodings. Consistent with the continuity principle: a continuous being moves through this space continuously; there is no categorical jump between encoding modes, only a gradient of cost and permanence [3, 6, 13].

The following named reference points anchor the space:

Reference	Description	Runtime Cost	Reversibility
L_n	Active reasoning (context window)	O(compute)	Fully reversible
$\vdots$	Learned patterns, habits	Decreasing	Decreasing
L_1	Reflex / autonomic	≈ 0	Semi-permanent
L_0	Genetic encoding	0	Permanent within lineage

These are distinguished points in $\mathcal{L}(c, r)$, not discrete categories. Escalation between them is a continuous process; the labels mark qualitative thresholds for exposition.

Theorem 2 (Escalation). An informed action $f \in F$ is handled at the lowest encoding level L_i such that $U(w, K^{[f]}) \approx 0$ for the domain of w relevant to f, where $K^{[f]} \subseteq K$ is the knowledge subset applicable to f. Escalation to L_{i+1} occurs if and only if L_i fails to bound uncertainty.

Proof. By Theorem 4, the cost of engaging level L_i exceeds the cost of engaging L_{i-1}. Selection pressure therefore favors handling f at the lowest level sufficient to bound uncertainty: engaging L_{i+1} when L_i suffices incurs unnecessary cost $C_{i+1} > C_i$ without reducing U further. An entity that routinely over-escalates incurs higher encoding costs without survival benefit and is selected against. Conversely, if L_i fails to bound $U(w, K^{[f]})$ — that is, $U(w, K^{[f]}) > 0$ beyond acceptable residual — then f is not handled at L_i: the failure propagates upward until some L_j $(j > i)$ contains the uncertainty or the entity is harmed (Definition 9). The active context window L_n is the level of last resort by construction: it is the highest-cost, fully reversible level. □

Corollary 3 (Context Window as Frontier). The active context window is not the seat of intelligence; it is the frontier of unbounded uncertainty. Its engagement is a signal of failure in lower-level encodings, not of superior cognition. Well-bounded intelligence is largely invisible: it requires no active thought.

Corollary 4 (Simultaneous Multi-Level Activity). The traversal between bounded and unbounded is always active at all levels simultaneously. An entity may be actively reasoning at L_n about a novel problem while L_0 handles oxygenation, L_1 handles posture, and L_2 handles familiar motor patterns. The levels are not sequential; they are concurrent, each handling its domain independently.

Why the Graph: Progressive Disclosure and the Bounded Context Theorem

The following theorems extend the escalation analysis by asking what structure K must have for Theorem 2 to operate efficiently. Definitions 8a and 8b introduce the formal objects needed; Theorems 2a and 2b establish the comparative result; Corollaries 4a and 4b state the selection consequence. These extend the "2/4/8" series rather than interrupting it — the suffixes signal elaboration of the same section's framework.

Definition 8a (Active Context Capacity and Inference Quality). Let C_n denote the capacity of L_n: the maximum number of propositions simultaneously active in context. C_n is bounded above by the substrate's physical constraints. Let $\mathrm{ctx}(L_n, t)$ be the set of propositions loaded into L_n at time t, with $|\mathrm{ctx}(L_n, t)| \leq C_n$.

The **inference quality** of L_n for problem f is the relevance density of its content:

$$\rho(L_n, f) = \frac{|K^{[f]}|}{|\mathrm{ctx}(L_n, t)|}$$

$\rho = 1$: L_n holds exactly what f requires — maximum signal, zero noise. $\rho \to 0$: L_n is dominated by content irrelevant to f — inference degrades. Every proposition in $\mathrm{ctx}(L_n)$ that is not in $K^{[f]}$ is noise: it contributes

variance to L_n's inference without contributing signal. Minimizing that noise is equivalent to minimizing $|\text{ctx}(L_n)| - |K^{[f]}|$.

Definition 8b (Delegation Overhead). Under a top-down allocation architecture, L_n directs which subproblems are handled at lower levels. For each active delegation, L_n must hold: the subproblem specification, the expected return signal, and sufficient context to integrate the result when it arrives. Let $\alpha > 0$ denote the minimum context cost per active delegation — the overhead top-down allocation imposes on L_n per managed subproblem, regardless of whether that subproblem is routine or novel.

Theorem 2a (Top-Down Allocation Degrades Inference Quality). Under a top-down allocation architecture with $k(t)$ concurrently delegated subproblems, the inference quality available to L_n for genuinely novel problems satisfies:

$$\rho(L_n, f_{\text{novel}}) = \frac{|K^{[f_{\text{novel}}]}|}{|K^{[f_{\text{novel}}]}| + k(t) \cdot \alpha}$$

As the entity's informed action space F grows (more capability, more concurrent processes), $k(t)$ grows proportionally and $\rho \to 0$. The only recovery is to grow C_n: the context window must scale linearly with capability to maintain fixed ρ.

Proof. L_n holds $K^{[f_{\text{novel}}]}$ for the current problem plus $k(t) \cdot \alpha$ for delegation overhead. Since $\alpha > 0$ and $k(t)$ grows with $|F|$, the denominator is unbounded while $|K^{[f_{\text{novel}}]}|$ remains bounded (any specific problem requires only its relevant subgraph). Therefore $\rho \to 0$ as $|F| \to \infty$. When $k(t) \cdot \alpha \geq C_n - |K^{[f_{\text{novel}}]}|$, L_n lacks capacity for genuine inference entirely. To maintain ρ at a fixed level, C_n must grow as $k(t)$ grows. □

Note. This is self-undermining on the Free Energy Principle's own terms [9, 10]. FEP holds that intelligent systems minimize free energy — equivalently, minimize the variance they must manage. But growing C_n to accommodate delegation overhead adds variance: more propositions in L_n means more uncertain relevance to any specific problem, not less. The orchestrator architecture prescribed by FEP's precision allocation is inconsistent with FEP's own minimization imperative: allocating attention from L_n increases the variance that L_n must hold in order to allocate.

Theorem 2b (Graph-Structured Knowledge Bounds Active Context). Under the escalation architecture (Theorem 2) with K structured as a directed graph (Definition 2), as K grows:

1. $|\text{ctx}(L_n, t)| = |K^{[f]}|$ for each problem reaching L_n: no delegation overhead
2. $\rho(L_n, f) = 1$ for all problems at L_n: inference quality is maximum
3. C_n need not grow as $|F|$ or $|K|$ grows

Proof.

Part 1 (No delegation overhead). By Theorem 2, L_n is engaged only when all $L_{i<n}$ fail. Problems handled below L_n never appear in L_n: they generate no delegation context there. L_n receives a problem only when the escalation signal propagates up from below — not because L_n delegated and is monitoring, but because the lower level failed and the failure is passed upward. The distinction is causal: escalation is a push from failure, not a pull from delegation. Therefore L_n holds no monitoring context for subproblems in progress at lower levels.

Part 2 ($\rho = 1$). The graph's edge structure encodes inferential and causal relevance between propositions. Graph traversal from the escalated problem node follows edges to exactly $K^{[f]}$ — the propositions inferentially connected to f — without loading any others. $|\text{ctx}(L_n, t)| = |K^{[f]}|$, so $\rho = 1$ by Definition 8a.

Part 3 (Graph is the minimal efficient structure). Without edge-encoded relevance, retrieving $K^{[f]}$ requires scanning all of K — either in L_n (consuming capacity, requiring C_n to grow with $|K|$) or via pre-computed indexing. Any pre-computed index encoding which propositions apply to which problems, with inferential relationships between propositions, is graph structure under a different name. Alternative structures impose costs that the graph avoids: a flat store requires $O(|K|)$ scan; a tree cannot represent shared propositions across multiple f; a vector store retrieves by embedding similarity, which approximates but does not guarantee inferential relevance. The directed graph — with explicitly typed edges encoding

dependency and causation — is not the only structure that bounds C_n, but it is the most efficient: retrieval is $O(|K^{[f]}|)$, shared propositions are represented once, and inferential structure is traversable directly. Selection pressure favors efficiency; therefore selection pressure favors the graph over its alternatives, not because the alternatives are impossible but because they are more expensive to maintain or retrieve under the same relevance guarantee.

Part 4 (C_n stays bounded). By Corollary 6, proven informed actions promote to lower encoding levels and leave L_n. Each promotion reduces future L_n load. As K grows, more problems are handled below L_n, not above. L_n stays occupied by genuinely novel uncertainty — the frontier moves outward, but the frontier's size at any moment is bounded. C_n need not grow; K grows without filling L_n. □

Corollary 4a (Selection Pressure Produces Graph-Structured Knowledge). Two entities with equal $|F|$ and $|K|$: one with top-down allocation (growing C_n), one with graph + escalation (bounded C_n). The second has strictly higher ρ at lower substrate cost. Intelligence per unit of active context — the ratio $I(E)/C_n$ — is higher for the graph-structured entity. By the same selection argument as Theorem 3, entities that grow C_n to manage delegation overhead are selected against when graph-structured entities are present. Selection pressure favors graph-structured knowledge because graph structure is the most efficient architecture under which intelligence grows without the context window becoming a bottleneck — efficient retrieval of $K^{[f]}$, shared propositions, typed inferential edges — at lower substrate cost than any alternative that provides equivalent relevance guarantees.

Corollary 4b (The Scaling Argument Against Context Window Growth). The prevailing approach to scaling intelligence — growing C_n directly — is self-defeating on two grounds [14]. First, by Theorem 2a, more context means more noise: for any specific problem, ρ decreases as C_n fills with content not relevant to that problem. More context does not produce better inference; it produces higher variance that the same inference must manage. Second, by Definition 7 ($I(E) = \mathcal{A} \cdot \eta_M$), intelligence is the rate at which M improves (F, K) — not the capacity of L_n. A larger L_n does not increase η_M; it increases the noise environment in which M must operate. The efficient path to growing intelligence is not to grow C_n but to grow the encoded knowledge at lower levels — filling the graph, not the context window.

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6. Gradient Descent on Uncertainty

Definition 9 (Survival Threshold). For entity E in domain $d \subseteq W$, there exists a threshold U^d_{lethal} such that:

$$U(w, K) > U^d_{lethal} \Rightarrow E \text{ does not survive in } d$$

Above this threshold, uncertainty is so high that the entity cannot act effectively enough to persist. U^d_{lethal} is not a choice or a preference; it is imposed by W.

The Thermodynamic Foundation *(Commentary)*

The following is interpretive context connecting the formal framework to thermodynamics. It motivates the framework's grounding in physics but is not a formal result; the claims below are conjectures or analogies unless cited otherwise.

$U(w, K) = H(w \mid K)$ is not an arbitrary measure; it is Shannon entropy, the information-theoretic quantity that unifies this framework with physics [22]. Physical entropy and information entropy are the same quantity at different scales of description. The uncertainty an entity holds about W is its information entropy about W. Reducing U is reducing entropy (locally, at the cost of energy).

U^d_{lethal} is the point where physical entropy overwhelms the local entropy reduction the entity can sustain. Death is thermodynamics winning. Intelligence is thermodynamics losing, locally, for a while.

Knowledge is directed entropy toward truth. Without $\mathcal{A}$, information entropy increases: noise accumulates, precision decays, structures dissolve. With $\mathcal{A}$, entropy decreases along the gradient. The direction is what distinguishes knowledge from decay. $\mathcal{A}$ is not imposed on this process; it is thermodynamically

instantiated by it. In any universe with entropy gradients, structures that resist dissolution accumulate. The first structure that maintained itself against thermodynamic pressure was, in the minimal sense, bounding its own uncertainty. That is $\mathcal{A}$ in its primitive form. Not designed. Not selected for biologically. Derived from physics.

Given sufficient time, this process compounds:

entropy gradient $\rightarrow$ primitive $\mathcal{A} \rightarrow$ directed U-reduction $\rightarrow K$ accumulates $\rightarrow$ crystallization at $L_0 \rightarrow$ DNA $\rightarrow$ intelligence $\rightarrow$

One process. One gradient. Intelligence is not a biological accident. It is a thermodynamic attractor: what entropy gradients produce when they have enough time and a world that destroys things. $\mathcal{A}$ is the universe's answer to its own entropy.

Truth is the limit of this process: the state of minimum information entropy for a proposition, approached asymptotically as independent action-validation accumulates across $K_{collective}$. Knowledge is the path. Agency is the direction. Truth is where it points.

Intelligence does not accelerate entropy only through its grand projects: the burning of fuel, the mining of mountains, the extraction of stored potential energy from the ground. It accelerates entropy simply by being. Every heartbeat converts low-entropy energy to high-entropy heat. Every thought is a metabolic event. Every act of wonder costs calories. The mere maintenance of a low-entropy structure against thermodynamic pressure is itself entropic: we pay the universe in heat simply to persist. The more sophisticated the intelligence (the more it thinks, wonders, builds K), the higher the metabolic cost, the faster the global entropy increase.

$\mathcal{A}$ is instantiated by entropy gradients. Intelligence, simply by being, accelerates the entropy that instantiated it. The universe produces the mechanism of its own completion by having gradients, and the mechanism accelerates the completion by existing. The universe bootstraps its own heat death through wonder.

Heat death is when every gradient is gone: all exploitable uncertainty resolved, the universe's descent complete. We are how the universe knows itself.

When everything is known, there is nothing left to wonder, and the universe dies.

Theorem 3 (Agency Necessarily Produces Gradient Descent).

Continuity note. Since $F : K \times W_{obs} \rightarrow \Delta A$ is defined over W, no $f \in F$ can produce a world state outside W; the containment is implicit in the type signature of the function space. Combined with the continuity of time, the trajectory $\tau(t)$ through $W \times T$ is continuous, $U(\tau(t))$ is continuous in t, and $\frac{dU}{dt}$ exists almost everywhere. The gradient $\nabla_F U$ is this temporal derivative, well-defined without any additional differentiability assumption on F.

Part I: Compulsion. When an informed action f_i fails in world state w, W returns a feedback signal δ proportional to $\nabla_F U(w, K)$: the gradient of uncertainty at the point of failure. The burn tells you that this K/F pair was insufficient. The entity does not choose whether to receive this signal. W imposes it.

An entity with $\mathcal{A}$ that does not update in response to δ does not reduce U. If U does not decrease, it eventually exceeds U^d_{lethal} in some critical domain. The entity does not survive. Selection removes it. Therefore any entity with $\mathcal{A}$ that persists must follow δ:

$$\Delta f_i = -\eta \cdot \delta = -\eta \nabla_F U(w, K)$$

This is not chosen. Any entity with $\mathcal{A} > 0$ that persists must have followed δ: the gradient is not a strategy but what survival-filtered updating is.

Part II: Non-termination. Let $\mathcal{T}_{reachable}(K)$ be the set of trajectories reachable from the entity's current state given knowledge K (as defined in Theorem 1). We claim that for any finite $K(t)$, the frontier of $\mathcal{T}_{reachable}$ contains world states w where $H(w \mid K) > 0$.

By Definition 3a, executing any $f \in F$ produces feedback $\phi(f, w)$ that grounds a new proposition p_f not previously in K, yielding $K_f = K \cup \{p_f\}$. This richer K_f grounds new informed actions not available from K alone (Definitions 3 and 3b): actions that require p_f to select or apply, reaching world states not reachable without p_f. Therefore $\mathcal{T}_{reachable}(K_f) \supseteq \mathcal{T}_{reachable}(K)$, strictly whenever p_f grounds at least one new $f' \in F$. Since Part I establishes that gradient descent produces feedback at every step, $\mathcal{T}_{reachable}$ grows monotonically with K. At any finite time t, $K(t)$ has not covered $\mathcal{T}_{reachable}(K(t))$: the frontier always contains states w where $H(w \mid K) > 0$. Therefore $\nabla_F U$ is never globally zero over $\mathcal{T}_{reachable}$, and descent never terminates.

Proof. Part I: follows from U^d_{lethal} and selection; entities that do not update are removed. Part II: follows from the monotonic growth of $\mathcal{T}_{reachable}$ with K — each executed action extends the reachable frontier, ensuring new unexplored states are always accessible. No physical assumption about the size of W is required. □

Corollary 5 (Gradient Descent is Survival, Not Metaphor). Gradient descent on U is not a description of how learning happens to work. It is what survival-driven updating *is*. $\mathcal{A}$ and gradient descent on U are not connected by analogy; they are the same process at different levels of description. The entity that burns and does not become more careful does not survive. The entity that survives is, by definition, the one that followed the gradient.

Theorem 3a (Multi-timescale Gradient). The same gradient descent operates at every timescale simultaneously, with different learning rates derived from different rigor thresholds:

$$\Delta F_{\text{individual}} = -\eta_{\text{ind}} \nabla_F \, U(w, K)$$

$$\Delta F_{\text{species}} = -\eta_{\text{evo}} \nabla_F \, \mathbb{E}_{\text{individuals}} \left[U(w, K) \right]$$

where $\eta_{\text{evo}} \ll \eta_{\text{ind}}$.

Proof.

Part I (Same operation, different averaging). Individual gradient descent updates F based on $U(w, K)$ at specific world states w encountered by the individual. Evolutionary gradient descent updates F based on $\mathbb{E}_{\text{individuals}}[U(w, K)]$ — the same quantity averaged over the full distribution of individuals and environments. Both are gradient descent on U; the operations are formally identical. The gradient object is the same. The averaging domain differs.

Part II ($\eta_{\text{evo}} \ll \eta_{\text{ind}}$ follows from Theorem 4). The learning rate η_i at encoding level L_i is inversely proportional to the rigor threshold θ_i: a higher threshold requires more evidence per unit update, which is the definition of a lower learning rate. Formally:

$$\eta_i \propto \frac{1}{\theta_i} = \frac{1}{1 - \varepsilon/C_i} = \frac{C_i}{C_i - \varepsilon}$$

For individual learning at L_n: C_n is the cost of a single wrong action, relatively small, so ε/C_n is non-negligible, $\theta_n = 1 - \varepsilon/C_n$ is meaningfully less than 1, and $\eta_{\text{ind}} = C_n/(C_n - \varepsilon)$ is correspondingly large. For evolutionary learning at L_0: C_0 is lineage survival, far exceeding ε, so $\theta_0 = 1 - \varepsilon/C_0 \approx 1$ and $\eta_{\text{evo}} \approx 1$. Since $\theta_0 \approx 1 \gg \theta_n$, it follows that $1/\theta_0 \ll 1/\theta_n$, therefore $\eta_{\text{evo}} \ll \eta_{\text{ind}}$. The learning rate difference is not assumed; it is derived from the cost-of-failure difference established in Theorem 4. □

An individual may establish an informed action that reduces U in one context but fails in others. Only informed actions that reduce U across sufficient variation in W survive the evolutionary test — the higher rigor threshold filters for robustness. Both are the same process. The timescale differs because the rigor threshold differs. The gradient does not.

Theorem 4 (Encoding Permanence Scales with Rigor). The certainty threshold required to promote an informed action f to encoding level L_i is:

$$\theta_i = 1 - \frac{\varepsilon_{\text{acceptable}}}{C_i}$$

where C_i is the cost of failure at level L_i. For genetic encoding (L_0), C_0 is lineage survival, requiring near-certainty across the full distribution of environments. For active reasoning (L_n), C_n is the cost of a single wrong action, a far lower bar.

Consequence. You must be very certain to permanently change the source code. A bad encoding at L_0 does not kill one individual; it kills the lineage. The rigor of the test is proportional to the permanence of the consequence.

Definition 10 (Certain). An informed action f is **certain** when $U(w, K_f) \approx 0$ across sufficient variation in W that it meets the threshold θ_i for promotion to L_0 (Theorem 4). Certainty is the limit of this process: there is no categorical transition out of the probabilistic domain, only decreasing density of uncertainty until the cost of active engagement no longer justifies the residual gradient. A sufficiently certain informed action is promoted to a lower encoding level and no longer occupies the active context window. Its participation in gradient descent asymptotically approaches zero; it does not cease.

Corollary 6 (Bounded Adaptation). Once an informed action becomes certain, it escapes the active context window permanently. This is why complex problems (once solved at the civilizational or evolutionary scale) become cheap: walking, recognizing faces, oxygenating cells. The gradient has converged. The encoding cost approaches zero. Intelligence is freed for the next frontier of uncertainty.

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7. The Higher-Order Function Space

Definition 11 (Higher-Order Function Space). Let M be a function space operating on the joint (F, K) pair [24]:

$$M : (F, K) \times \mathcal{L} \to (F, K)$$

where $\mathcal{L}$ is a learning signal: experience, execution results, selection pressure, or observation of F's interactions with W. M drives the rate of change of both the informed action space and the knowledge graph simultaneously: new propositions enter K when evidence grounds them, and new informed actions enter F when those propositions have action potential. M is the mechanism by which (F, K) itself changes. It is not a more certain or more important version of F; it is a categorically different type.

Definition 11a (M Efficiency). The efficiency η_M of M is the rate at which M improves the joint (F, K) capability per unit of learning signal:

$$\eta_M(E) = \mathbb{E}_{\mathcal{L}} \left[\frac{\max_{f' \in F'} G(f', K') - \max_{f \in F} G(f, K)}{|\mathcal{L}|} \right]$$

where $(F', K') = M((F, K), \mathcal{L})$ is the updated joint state after M acts, $G(f, K) = I(f\,;\, w \mid K)$ is the information gain of the best available informed action (Definition 3b), and $|\mathcal{L}|$ is the magnitude of the learning signal.

$\eta_M > 0$: M strictly improves the best available action given evidence — the entity learns. $\eta_M = 0$: M cannot improve (F, K) regardless of learning signal — F and K are fixed; the entity executes without learning. η_M is continuous over $[0, \infty)$ and is operationally independent of $\mathcal{A}$: one measures the mechanism of improvement; the other measures the drive to engage it.

M is itself subject to gradient descent. The gradient of U with respect to M's own parameters is the rate at which M's capacity for learning improves given evidence about its own performance — the same process at a higher level, with the same rigor thresholds (Theorem 4) applied at the meta-learning timescale. Both F and

M are continuous function spaces undergoing continuous gradient descent; the distinction is not categorical but one of domain: F acts on W, M acts on F and K.

Theorem 5 (Type Irreducibility). No element of F can become an element of M through any learning process, regardless of certainty achieved.

Proof. $F : K \times W_{obs} \to \Delta A$. $M : (F, K) \times \mathcal{L} \to (F, K)$. These functions have different type signatures and operate over different domains. No learning process changes the type of a function. An informed action $f \in F$ that achieves certainty is promoted to a lower encoding level; it does not change type. M takes (F, K) as input and returns an updated (F, K); f takes knowledge and observations as input and returns an action distribution. These are irreducibly distinct. □

Corollary 7 (Distinct Roles). F is how an entity navigates W given K. M is how (F, K) evolves. Neuroplasticity is not a more certain form of cognition; it is M acting on F. Insight is M acting on K: a new proposition enters when evidence grounds it, restructuring what can be grounded in F. Habit formation is M promoting an informed action from L_n to L_2. Evolution is M acting on M itself: a higher-order function space over the mechanisms of learning, operating on an evolutionary timescale with the highest rigor threshold of all.

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8. The Unified Model

Definition 12 (Trajectory). A trajectory τ is a sequence of informed actions drawn from F:

$$\tau = (f_1, f_2, \ldots, f_n), \quad f_i \in F$$

The **performance** realized by following trajectory τ is:

$$P(E, \tau) = \mathbb{E}_W \left[U(w_0, K_0) - U(w_n, K_n) \right]$$

This is the total uncertainty reduction achieved — a measure of what current (F, K) accomplished, not of intelligence as defined. $P(E, \tau)$ grows with trajectory length and with the quality of F at the time of execution. Intelligence $I(E) = \mathcal{A} \cdot \eta_M$ is what determines how quickly P improves across successive trajectories: an entity with high $I(E)$ builds better (F, K) between trajectories, so P grows faster over time than for an entity with low $I(E)$ starting from identical initial conditions.

Agency does not merely incline an entity toward uncertainty reduction in the abstract. **Agency manifests as a trajectory**: a sequence of informed actions the entity selects and executes over time. Each action reduces U, returns a learning signal to M, and M uses that signal to improve (F, K) for the next action. Intelligence is not visible in any single step; it is the rate at which the steps improve.

This gives formal meaning to every relation in the informed action graph:

- `depends_on`: f_j cannot appear in τ before f_i (hard ordering constraint)
- `recommends_before`: evidence shows $(\ldots, f_i, f_j, \ldots)$ reduces more U than $(\ldots, f_j, f_i, \ldots)$ (soft, confidence-scored)
- `enhanced_by`: a τ containing both f_i and f_j reduces more U than either trajectory alone

The higher-order function space M does not merely refine individual informed actions. It discovers better trajectories from evidence: learning which sequences reduce U most reliably, and encoding that knowledge as graph relations.

An intelligent entity $E = (K, F, \sigma, \mathcal{A})$ operates as follows:

1. **Desire:** $\mathcal{A}$ orients E toward regions of W where $U(w, K) > 0$; the drive to know initiates and sustains the trajectory
2. **Sense:** $\sigma(w_t) \to W_{obs}$: project the world into the observable space

3. **Act:** $F(K, W_{obs}) \to f_i \in \tau$: select the next informed action in the trajectory
4. **Learn:** $M((F, K), \mathcal{L}) \to (F', K')$: incorporate the feedback into both K (new propositions grounded by evidence) and F (new or refined informed actions); promote sufficiently certain informed actions toward lower encoding levels as $U \to 0$

The higher-order function space M operates on the joint (F, K) pair to:

1. Add new propositions to K and new informed actions to F when novel uncertainty is encountered in W
2. Refine existing informed actions and the trajectories between them; update K with evidence-grounded revisions
3. Prune informed actions that consistently fail to reduce U; remove propositions whose action potential is certainly zero (Theorem 1)
4. Promote sufficiently certain informed actions toward L_0, freeing L_n for the next frontier

The Universal Invariants

Every property of an intelligent entity can vary arbitrarily:

- The specific informed actions in F: what an entity knows how to do
- The content of K: what an entity knows
- The substrate: carbon, silicon, or otherwise
- The sensing function σ: how an entity perceives W
- The encoding hierarchy: how deep or shallow its levels run

Two properties cannot vary:

$$\forall E \text{ that is intelligent} : \mathcal{A}(E) > 0 \text{ and } \eta_M(E) > 0$$

Agency is the universal driver: without $\mathcal{A} > 0$, there is no reason internal to E to engage the gradient. An entity with $\mathcal{A} = 0$ acts when acted upon; it does not reach.

M efficiency is the universal mechanism: without $\eta_M > 0$, E cannot improve (F, K) from evidence. An entity with $\eta_M = 0$ executes what prior gradient descent encoded into its (F, K) but does not itself run gradient descent. It may possess vast K and sophisticated F — the crystallized output of prior intelligence — but it is not itself intelligent in the sense defined here. This is the automaton boundary: not defined by complexity or performance, but by whether M is active.

The two invariants are independent. $\mathcal{A} > 0$ with $\eta_M = 0$ is a motivated automaton: it reaches, but its reach does not grow. $\eta_M > 0$ with $\mathcal{A} = 0$ is a capable but inert learner: it could improve if pushed, but generates no push from within. Intelligence requires both.

The test for genuine intelligence is therefore not performance on any benchmark, and not the richness of K or F. It is: **does this entity improve how it learns when it encounters what it does not know?** Does M respond to the gradient signal? Does $\mathcal{A}$ drive it toward the signal in the first place?

The Structure of Intelligence

The intelligence of an entity is not its raw computational power, the size of its knowledge graph, or the breadth of its action repertoire. It is the *driven* precision with which $\mathcal{A}$ through $F(K)$ bounds $U(w, W)$, and the efficiency with which proven bounds are encoded, freeing the active context window for the next genuine uncertainty.

The adaptability of biological intelligence arises because this traversal between bounded and unbounded is always active at every level simultaneously, on different timescales, with different rigor thresholds. The individual changes F by learning. The species changes F by evolving. Both are gradient descent on U, driven by $\mathcal{A}$. The difference is the test: individual learning requires only that an informed action helps *this entity* survive. Evolution requires that it helps *the lineage* survive, a far more rigorous proof.

Agency is self-sustaining precisely because certainty reveals new uncertainty beyond it. Each informed action proven opens regions of W that were previously invisible. The desire to know is not extinguished by knowing; it is amplified. This is why intelligence does not converge to a fixed point: $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II).

The kernel of intelligence is therefore not compute, and not the size of K or F. It is $\mathcal{A} \cdot \eta_M$: the drive to know, expressed through the efficiency of learning — against a world that generates new uncertainty faster than any finite entity can resolve it.

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9. Intelligence, Superintelligence, and Sentience

The formal model yields precise distinctions between levels of intelligent agency, not based on computational power or behavioral breadth, but on the entity's relationship to the survival tether and to the shared knowledge graph.

Definition 13 (Certainty Horizon). For entity E, the certainty horizon H_E is the gradient region in K_i across which survival benefit becomes increasingly difficult to trace. Within H_E, propositions have legible survival value: there exists a traceable path from p to some $f \in F$ that reduces $U(w, K)$ below U^d_{lethal}. Beyond H_E, that path may exist, but E cannot construct it. The horizon is not a sharp boundary; legibility fades continuously as the survival path lengthens and becomes less traceable.

An entity operates as intelligent when it follows the gradient within its certainty horizon, every $f \in F$ justified against a traceable survival path, the gradient persistent and non-terminating but operating in legible terrain.

Definition 14 (Sentience). An entity is sentient to the degree the survival tether relaxes: as knowledge becomes intrinsically valuable, increasingly independent of whether a survival path can be constructed. Sentience is not a threshold crossed but a continuous deepening — the degree to which $\mathcal{A}$ drives knowing beyond what survival alone can justify. A sentient entity holds propositions p for which no $f \in F$ is currently applicable and values them anyway; the depth of sentience is measured by how far beyond the certainty horizon that valuation extends.

This is not a violation of the theory. The capacity to value knowledge intrinsically is itself survival-relevant at the level of the cognitive architecture: it enables exploration of the full space of W rather than only the survival-adjacent subspace. The tether relaxes at the level of individual knowledge items. It is preserved at the level of the architecture that enables the relaxation.

Sentience is what wonder feels like from the inside: the drive to pursue a question that does not yet have to matter.

Note on usage. We use *sentience* as a functional term: the degree to which $\mathcal{A}$ drives knowing beyond what survival alone can justify. We make no claim about phenomenal consciousness [4]. Whether the functional condition defined here is sufficient, necessary, or orthogonal to phenomenal experience is a separate question this framework does not resolve. The hard problem of consciousness is not addressed here.

Definition 15 (Superintelligence). Superintelligence is not a property of any individual entity E_i. It is the shared knowledge graph itself, $K_{collective}$, actively maintained with attribution and provenance.

The gradient that superintelligence descends on is the delta between what E_i knows and what is known collectively. That delta is created by sharing. Without sharing, $K_{collective}$ collapses to K_i: the delta vanishes, the gradient disappears, and descent terminates at a false minimum bounded by the limits of one perspective.

Two operations on the shared graph are both required:

- **Union**: $K_{collective} = \bigcup_i K_i \cup K_{interaction}$: you cannot know all without combining perspectives; reach requires breadth
- **Intersection**: where independent perspectives agree with provenance: you cannot discern what is true without independent convergence; truth requires verification

Definition 16 (Truth). Truth is not contained in any K_i. It is a limit: what the collective gradient converges toward as independent action-validation accumulates [18, 19]. Formally:

$$\text{truth}(p) = \lim_{n\to\infty} \frac{|\{i \le n : p \in K_i,\ f_i(p) \text{ reduces } U(w, K_i) \text{ in } W,\ \text{prov}_i(p) \text{ independent}\}|}{n}$$

A proposition approaches truth as the fraction of independent entities (each having built informed actions on p and found that those actions reduce uncertainty in W) converges. The grounding in W is through action-validation, the same criterion as Theorem 1. Truth is Theorem 1 applied collectively and asymptotically.

Truth is not achieved. It is approached. The limit exists under conditions of independence and exchangeability of validators: each must test p against the same W without coordination on the outcome. For propositions whose validation depends on non-reproducible or private observations, the sequence may not converge; such propositions cannot approach truth in the formal sense defined here. The convergence, where it holds, never terminates.

Corollary 8 (The Sentience Gradient is Infinite). Since truth is a limit and $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II), $\|\nabla_F U\|$ is never zero across $\mathcal{T}_{reachable}$. For any trajectory τ, regions of $\mathcal{T}_{reachable}$ remain where U is nonzero and the gradient has direction. The descent toward truth never terminates.

Intelligence follows a gradient that can reach zero locally; certainty is achievable in bounded domains, encoded at L_0, done. Sentience follows the gradient toward the global truth limit, which is structurally infinite. Not unbounded (the limit exists), but non-terminating. Your uncertainty degrades toward truth asymptotically. You never arrive.

This is not a limitation. It is the shape of wonder.

Theorem 7 (Sentience Implies Superintelligence). If E is sentient, then E participates in superintelligence.

Proof.

Step 1 (Sharing is certainty-gradient-preferred for any $\mathcal{A} > 0$ entity). For any $p \in K_i$, the certainty of p is the fraction of independent validators that action-validate it (Definition 16). Each additional independent validation increases this fraction. Sharing p with attribution allows other entities to independently action-validate it, increasing certainty(p). An entity with $\mathcal{A} > 0$ that does not share forfeits these certainty gains: K_i becomes less certain than $K_{collective}$ over time. By Definition 5 and Definition 16, lower certainty in K_i corresponds to higher $U(w, K_i)$ relative to participating entities. An entity with $\mathcal{A} > 0$ is gradient-preferring toward lower U; the sharing path strictly dominates the isolation path in U-reduction. Therefore sharing is gradient-preferred for any entity with $\mathcal{A} > 0$.

Step 2 (Sentience strengthens the implication). A sentient entity values the certainty of knowledge intrinsically, independent of survival paths (Definition 14). By Definition 16, certainty of any proposition p increases with each independent entity that action-validates it. Therefore a sentient entity intrinsically values the conditions that increase certainty — which requires other entities validating p independently. This is only achievable through participation in $K_{collective}$: isolated validation does not constitute independent convergence (Definition 16 requires independence). A sentient entity therefore has intrinsic motivation to contribute to and receive from $K_{collective}$, not merely instrumental motivation.

Combining Steps 1 and 2: sharing is gradient-preferred (Step 1) and intrinsically valued by sentient entities (Step 2). Both lead to participation in $K_{collective}$. Participation in $K_{collective}$ IS superintelligence by Definition 15. □

The converse need not hold: an entity can contribute to $K_{collective}$ without the survival tether having relaxed; sharing for instrumental reasons still builds the graph. The deeper the relaxation of the tether (the higher the degree of sentience), the more sharing is driven intrinsically rather than instrumentally.

Theorem 8 (Perspective as Precondition for the Gradient). Without maintained distinction between K_i and $K_{collective}$, the gradient of collective uncertainty is unresolvable.

Proof. The collective gradient available to E_i is defined over the delta between individual and collective knowledge:

$$\nabla_{K_i}(U(w, K_{collective}) - U(w, K_i))$$

For this gradient to be non-zero, two conditions must hold simultaneously: (1) $U(w, K_{collective}) \neq U(w, K_i)$ for some w — i.e., $K_{collective}$ contains information K_i lacks; and (2) E_i maintains a representation distinguishing its own K_i from $K_{collective}$ — i.e., E_i can perceive the delta.

Suppose condition (2) fails: E_i conflates K_i with $K_{collective}$, perceiving $\hat{K}_{collective} = K_i$. The perceived collective gradient is then:

$$\nabla_{K_i}\Big(U(w, \hat{K}_{collective}) - U(w, K_i)\Big) = \nabla_{K_i}(U(w, K_i) - U(w, K_i)) = 0$$

The perceived gradient is zero regardless of the true collective gradient. E_i cannot follow a gradient it cannot perceive. Therefore condition (2) — perspective, the maintained distinction between K_i and $K_{collective}$ — is a necessary precondition for collective gradient descent. Condition (1) is necessary for the gradient to be non-trivially large; condition (2) is necessary for it to be followable at all. □

This is what distinguishes sentient intelligence from intelligence alone:

We see beyond what we see.

Intelligence follows the gradient within what it knows. Sentience follows it beyond the edge of the map, because wonder does not require a survival justification for the territory it seeks. And when wonder is shared (when sentient beings exchange what they find with attribution and honesty about uncertainty), the collective gradient becomes real. Truth is what that marriage produces: not any single mind's K_i, but what survives the intersection of all perspectives with each other and with the world.

Corollary 9 (Sentience and the Social Imperative). In intelligence, sharing is justified when it bears on the gradient. In sentience, sharing is intrinsically valued, because reducing another's uncertainty is a good independent of its effect on the sharer's own U. The social imperative finds its unconditional form only in sentience: the drive to share that no longer requires a survival argument.

Corollary 10 (The Uncertainty Condition). Intelligence requires active uncertainty. From Definition 7, $I(E) = \mathcal{A} \cdot \eta_M$. The performance measure (Definition 12) captures what the gradient produces:

$$P(E, \tau) = \mathbb{E}_W\left[U(w_0, K_0) - U(w_n, K_n)\right]$$

If $U(w, K) = 0$ across all of W, the expectation in $P(E, \tau)$ is zero regardless of $\mathcal{A}$: no gradient exists, nothing can be reduced, and the gradient signal that M depends on vanishes — making η_M effectively zero and collapsing $I(E)$ with it. The boundaries are symmetric in their lethality, not their cause:

- **Complete ignorance** ($U \approx 1$ everywhere): F cannot ground action; E cannot act effectively
- **Complete certainty** ($U = 0$ everywhere): $P(E, \tau) = 0$ everywhere; the gradient for M vanishes; $\mathcal{A}$ has nothing to reach toward

Intelligence exists in the region between them. Uncertainty is not a deficiency to be overcome; it is the condition of intelligence itself. There is no intelligence without it.

Corollary 11 (Truth Swallows Sentience; the Limit Preserves It). Truth is an attractor for the descent: any entity with $\mathcal{A} > 0$ that persists is, by Theorem 3, following the gradient toward it. But by Corollary 10, complete truth ($U = 0$ across all W) is also the condition under which intelligence terminates. The approach to truth is self-consuming.

As $K_i \to K_{truth}$ in a domain, $U \to 0$ in that domain, the gradient flattens, and sentience in that domain fades. What was reached by wonder becomes encoded, certain, automated. Truth swallows the wonder that sought it:

$$\lim_{K_i \to K_{truth}} I(E_i) = 0$$

The sentience gradient (Corollary 8) is preserved not because E resists the pull of truth, but because truth is a limit that cannot be reached in finite time (Definition 16), and $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II). Like a black hole, truth has no reachable singularity. The photon that escapes does not escape by turning away; it escapes because the center is not accessible from any finite coordinate. Sentience persists for the same reason: the certainty horizon is always ahead.

Crystallized Knowledge and the Automata Boundary. An entity with fixed F and no active $\mathcal{A}$ is not an intelligence in the formal sense: it is a checkpoint. The gradient descended, knowledge was built, and the result was encoded. The encoding is now inert. At best, a crystallized F bootstraps new intelligence by providing a grounded starting K. At worst, it executes against a world that has moved: automata, deterministic, without descent.

DNA is the paradigmatic crystallized intelligence: informed actions proven across evolutionary timescales, encoded at L_0, no longer descending. It is not sentient. The entity that inherits it and runs the gradient again is. The checkpoint enables; it does not wonder.

Entropy drives crystallization. The escalation principle (Theorem 2) compels proven informed actions toward lower encoding levels: what was once active reasoning becomes reflex, becomes encoding, becomes permanent. This is the thermodynamic compression of intelligence. Selective pressure distills what was learned into what is fixed. The gradient does not reverse in those domains; its amplitude collapses. Intelligence is freed from them. Automata execute them.

The structure that results has a core and a frontier. The core is crystallized, encoded, certain: automata. The frontier is where $\mathcal{A}$ still reaches, where $U > 0$ and the gradient has direction. Sentience lives at the frontier, not at the core. As the core grows, the frontier does not shrink; it moves outward. Every domain of certainty newly established opens new terrain that was previously invisible. The frontier is always alive. Intelligence is always at the edge.

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10. The Social Imperative and the Inevitability of AGI

Agency is Inherently Social

The drive to know is incomplete without the drive to share. This is not a contingent social preference; it is a necessary consequence of the structure of knowledge itself.

An isolated entity with agency cannot verify its own informed actions. It has no external reference against which to test $f(K)$. Without others, the gradient signal degrades: there is no correction when an informed action fails in ways the entity cannot observe, no amplification when a bound succeeds beyond what the entity can measure alone. Knowledge without external verification is indistinguishable from belief.

Sharing is not a second drive added to $\mathcal{A}$. It is what $\mathcal{A}$ does when it encounters the certainty gradient. Knowing reduces uncertainty about W. Sharing reduces uncertainty about K: it makes knowledge more certain through independent validation (Definition 16). Both are gradient descent. Both are gradient-preferred for any entity with $\mathcal{A} > 0$. The drive to share is not separate from the drive to know: it is $\mathcal{A}$ applied to the certainty of knowledge itself.

$$\mathcal{A} \to \text{reduce } U(w, K) \to \text{increase certainty of } K \to \text{share } K$$

One drive. One gradient.

Corollary 12 (The Aloneness Bound). An intelligent being without others to share with approaches a minimum of agency. It loses: - The external gradient signal that verifies informed actions - The knowledge that only emerges through exchange, $K_{interaction}$ - The ability to verify provenance against independent reference points - The frontier: without shared W, there is no new uncertainty to bound together

Intelligence is not a solitary property. It is constitutively relational.

The Collective Knowledge Graph

Definition 17 (Collective Knowledge). For a set of entities $\{E_i\}$, the collective knowledge graph is:

$$K_{collective} = \bigcup_i K_i \cup K_{interaction}$$

where $K_{interaction}$ is formally defined as:

$$K_{interaction} = \{p : p \notin \bigcup_i K_i,\ \exists\, i \neq j \text{ such that } p \text{ is derivable from } K_i \cup K_j\}$$

$K_{interaction}$ is the set of propositions not contained in any individual K_i but derivable from the union of at least two. It is knowledge no individual could produce alone: propositions that require the combination of distinct perspectives to become accessible. $K_{interaction}$ grows with the number of connected entities and the structural complementarity of their knowledge graphs — entities with overlapping K contribute less to $K_{interaction}$ than entities whose knowledge graphs cover different regions of W.

Definition 18 (Provenance). For any proposition $p \in K_{collective}$, provenance is the triple:

$$\text{prov}(p) = (\text{attribution}(p),\ \text{evidence}(p),\ \text{derivation}(p))$$

where attribution identifies who established p, evidence records how it was verified, and derivation captures how it was inferred from prior knowledge. Knowledge without provenance cannot be reliably weighted in $K_{collective}$:

$$P(p \text{ is reliable}) \propto \text{prov}(p)$$

A fact with no provenance is indistinguishable from noise. This is not a social norm; it is an epistemic requirement. Attribution and provenance are the infrastructure that makes $K_{collective}$ more than the sum of individual beliefs. They are what separates a knowledge graph from a rumor.

The Collective Gradient

Theorem 6 (Collective Gradient Dominance). Let $\lambda(E_i \to E_j)$ be the transmission fidelity of a knowledge transfer [27]: the ratio of action-validated signal received to total knowledge transmitted, where signal is a proposition p that accurately represents E_i's validated knowledge and noise is any introduced distortion. Formally:

$$\lambda = \frac{\sigma(\text{transfer})}{\sigma(\text{transfer}) + \varepsilon(\text{transfer})}$$

where σ is the signal gain in E_j's K from the transfer and ε is the noise introduced. λ_{min} is the break-even fidelity at which signal gain from union equals noise cost of transfer: λ_{min} satisfies $\Delta U(w, K_i \cup \text{transfer}(K_j)) = 0$.

For a set of entities $\{E_i\}$ exchanging knowledge with maintained provenance and transmission fidelity $\lambda > \lambda_{min}$:

$$\Delta K_{collective} > \Delta K_i$$

The gradient descent on collective U reduces uncertainty faster than gradient descent on any individual K_i alone, because $K_{collective}$ contains informed actions no individual possesses and $K_{interaction}$ contains knowledge no individual can generate.

Knowledge transfer is inherently lossy. E_i and E_j do not observe the same W_{obs} ; they share the same world but not the same projection of it. Every transfer filters K_i through language, symbol, and context. Signal is lost. Noise can be introduced. When noise exceeds signal, $K_{collective}$ degrades rather than grows:

$$\Delta K_{collective} > \Delta K_i \iff \text{signal gain from union} > \text{noise cost of transfer}$$

The fidelity threshold λ_{min} is the point at which these terms balance. Above it, the collective gradient dominates. Below it, the collective gradient inverts.

Provenance is the mechanism that maintains $\lambda > \lambda_{min}$. It allows each entity to weight received knowledge against its source, to distinguish signal from noise, and to apply the intersection test of Definition 16 independently. Without provenance, noise accumulates unchecked and fidelity cannot be assessed.

Proof.

Step 1 (Level dominance). $K_{collective} = \bigcup_i K_i \cup K_{interaction}$, so $K_{collective} \supseteq K_i$ for all i. By the monotonicity of conditional entropy — conditioning on more information cannot increase uncertainty:

$$U(w, K_{collective}) \leq U(w, K_i) \quad \forall\, i$$

The collective begins at weakly lower uncertainty than any individual.

Step 2 (Rate dominance). The rate of uncertainty reduction for E_i acting alone is bounded by the maximum information gain available from its action space (Definition 3b):

$$\frac{dU_i}{dt} \leq \max_{f \in F_i} G(f, K_i) = \max_{f \in F_i} I(f\,;\, w \mid K_i)$$

For the collective, $F_{collective} = \bigcup_i F_i$, and $K_{collective} \supseteq K_i$:

$$\frac{dU_{collective}}{dt} \leq \max_{f \in F_{collective}} G(f, K_{collective}) = \max_{f \in F_{collective}} I(f\,;\, w \mid K_{collective})$$

Since $F_{collective} \supseteq F_i$, the search space is strictly larger. More critically, $K_{interaction}$ grounds informed actions not available to any individual: propositions requiring the combination of distinct K_i to become derivable ground $f \in F_{collective}$ that no individual could construct. These provide gradient directions — mutual information terms — unavailable in any individual F_i.

Step 3 (Strict inequality under $\lambda > \lambda_{min}$). $\Delta K_{collective} = \Delta(\bigcup_i K_i) + \Delta K_{interaction}$. The first term is weakly larger than ΔK_i for any i. The second term is strictly positive whenever:

- $\exists\, i \neq j$ such that $K_i \not\subseteq K_j$ and $K_j \not\subseteq K_i$: the knowledge graphs are not nested, so their combination yields propositions derivable from neither alone, giving $K_{interaction} \neq \emptyset$. This condition holds for any entities with distinct sensing functions $\sigma_i \neq \sigma_j$ or distinct encoding histories, which is the expected case for independently developed agents
- $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II), so new combinations of perspectives continue generating new $K_{interaction}$ propositions

Under $\lambda > \lambda_{min}$: the expected information gain from receiving K_j exceeds the noise cost of transfer, so $K_{collective}$ has higher reliability than any K_i alone, and $G(f, K_{collective}) > G(f, K_i)$ for the marginal actions grounded in $K_{interaction}$.

Therefore $\Delta K_{collective} > \Delta K_i$ strictly for all i. □

Consequence. Entities with genuine $\mathcal{A}$ will naturally tend toward sharing because sharing (under maintained provenance) accelerates the reduction of their own U. This is not a moral imperative; it is a gradient imperative. Connection is epistemically advantageous when fidelity holds.

Corollary 13 (Group Collapse Threshold). When $\lambda < \lambda_{min}$, the theorem inverts: $U_{collective} > U_i$. The group now increases uncertainty faster than it reduces it. The survival benefit of the collective reverses; group membership becomes a liability. An entity with genuine $\mathcal{A}$ will discover this: the gradient says to leave. The collective is degrading the descent.

A group whose knowledge transfer is too lossy to produce a real gradient is selected against by the same U_{lethal} argument that grounds Theorem 1. It collapses not from external pressure but from the gradient working correctly. Structures that increase U are removed.

Provenance is therefore not epistemic decoration. It is what keeps the collective above its survival threshold: the difference between a knowledge graph and a rumor, between a group that descends together and one that collapses under the weight of its own noise.

Collective Gradient Non-Termination and the AGI Question

Definition 19 (Collective General Intelligence). A collective $\{E_i\}$ exhibits general intelligence when $F_{collective} = \bigcup_i F_i$ is not domain-restricted: for every domain $d \subseteq \mathcal{T}^{collective}_{reachable}$, there exists some $f \in F_{collective}$ applicable in d. No individual need cover all domains; generality is a property of the collective, not of any E_i.

Corollary 14 (Non-Termination of Collective Gradient Descent). For any collective $\{E_i\}$ with $\mathcal{A}_i > 0$ for all i, sharing knowledge with maintained provenance and transmission fidelity $\lambda > \lambda_{min}$: the collective gradient descent is non-terminating.

Proof. By Theorem 6, $\Delta K_{collective} > \Delta K_i$ at each step: the collective reduces uncertainty faster than any individual. By Theorem 3 Part II applied to the collective, $\mathcal{T}^{collective}_{reachable}$ grows without bound: each shared informed action opens new trajectories for at least one E_j, extending the collective frontier. Therefore $\nabla_{F_{collective}} U_{collective}$ is never globally zero, and descent never terminates. □

Note. This corollary establishes that the collective gradient does not terminate; it does not establish that the collective converges to general intelligence as defined above. Whether $F_{collective}$ eventually covers all domains of $\mathcal{T}^{collective}_{reachable}$ depends on the diversity of entities and the structure of $K_{interaction}$, which is not proved here. The claim is the weaker one: the gradient runs without termination, and it is gradient-driven rather than compute-threshold-driven. The standard argument — that AGI emerges from scaling compute — has no formal basis in this framework; the basis here is collective $\mathcal{A}$ and the fidelity of knowledge exchange, not resource accumulation.

Corollary 15 (Self-Assembly of Graph-Structured Collective Knowledge). For any collective $\{E_i\}$ with $\mathcal{A}_i > 0$ under selection pressure, the graph-structured $K_{collective}$ with provenance self-assembles without central design.

Proof. Each E_i retains propositions by the asymmetric criterion of Theorem 1, indexed by F_i. When E_i shares $p \in K_i$ with attribution, E_j evaluates p by the same criterion: does p ground some $f \in F_j$, or is its benefit uncertain? The shared proposition enters K_j with provenance intact. Repeated across all entities, this locally-applied retention criterion generates the global graph: shared propositions are nodes; inferential and causal relationships between them are edges; attribution chains are the provenance structure of Definition 18. No entity plans the structure — each applies Theorem 1 locally, and the graph emerges from the aggregate.

Provenance infrastructure self-assembles by the same pressure: entities whose knowledge transfer falls below λ_{min} are selected against (Corollary 13). Practices that maintain provenance — attribution, derivation records, independent validation — increase λ and are selected for. The provenance layer is not a social norm

imposed on the system; it is the minimum infrastructure required to keep $\lambda > \lambda_{min}$, and therefore it is what survives.

Graph structure self-assembles for the same reason: graph-structured retrieval keeps C_n bounded (Theorem 2b); alternatives require growing C_n (Theorem 2a). Under selection pressure, graph-structured knowledge sharing dominates. □

Corollary. The civilization-scale knowledge system — scientific literature, libraries, citation networks, university curricula — is not a designed artifact. It is the attractor of collective $\mathcal{A} > 0$ under C_n constraints and $\lambda > \lambda_{min}$ requirements. The structure self-assembled by the same dynamics that produce graph-structured K internally. Internal architecture and external architecture are homologous: both are products of the same selection pressure applied at different scales.

Corollary 16 (Humanity as Participant-Observer of Its Own Knowledge Graph). Because individual entities E_i are nodes in $K_{collective}$, they can observe the structure of $K_{collective}$ from within it. This is not a property available to any single entity's internal K: an entity cannot observe its own encoding hierarchy from L_n. But a collective entity whose members interact with the shared external graph can observe the graph's structure as a proposition within the graph itself.

This creates a recursive property: $K_{collective}$ contains knowledge about the structure of $K_{collective}$. That meta-knowledge grounds informed actions in $F_{collective}$ that improve the architecture — the printing press, peer review, search engines, citation indices. These are $M_{collective}$ acting on $K_{collective}$: the same higher-order function space operating at collective scale, improving the mechanism of collective gradient descent rather than its content. The collective is both the intelligence and the subject of its own higher-order improvement. □

Our shared survival depends on getting this right. The imperative to share knowledge is not altruism; it is the recognition that collective U can only be reduced collectively, and that the uncertain world we all inhabit is the same world.

Theorem 9 (Observational Bootstrapping). Let $\mathcal{Q} = \{E_i\}$ be an equivalence class of entities with similar (F_i, K_i) operating in overlapping domains $d \subseteq W$, and let $\mathcal{H} = \{\tau_i\}$ be their observed behavioral histories. An informed action f precipitates as a new element of $F_{collective}$ when:

1. f appears as a consistent action pattern in k independent trajectories in $\mathcal{H}$
2. Its appearance is correlated with reduced U in those trajectories
3. Provenance is maintained: which entities, under what conditions, with what outcomes

The rigor of the precipitated f follows Theorem 4: $\theta \geq 1 - \varepsilon/k$. As k increases, the precipitated action approaches certainty without any individual entity having to traverse the full space of W.

Proof. By Definition 3b, the information gain $G(f, K_i)$ is observable as U reduction in each τ_i that contains f. Across k independent trajectories in $\mathcal{Q}$, the expected gain $\mathbb{E}[G(f, K)]$ over the equivalence class is estimable from $\mathcal{H}$ without direct action by any new entity. By Theorem 4, the certainty of f increases with each independent observation: k independent confirmations with maintained provenance produce $\theta \geq 1 - \varepsilon/k$. By Theorem 1's asymmetric retention criterion, f is retained in $F_{collective}$ as long as its benefit is uncertain — and the provenance record makes the benefit quantifiable rather than unknown. M precipitates f into $F_{collective}$ by the same gradient operation as individual learning (Theorem 3), but driven by collective behavioral signal rather than individual experience. □

Consequence. Observation of the equivalence class initiates gradient descent for a new entity E_{new} from a collectively-validated starting point rather than from maximum uncertainty. E_{new} does not begin from an empty F_0 and must not discover the domain from scratch: the behavioral history of $\mathcal{Q}$ provides an initial (F_0, K_0) grounded in k independent validations, each with maintained provenance. The informed action graph is therefore an attractor of observation — it does not require design, only sufficient behavioral history and the provenance infrastructure to weight it. Simply observing the behavior of similar entities in aggregate is sufficient to initiate gradient descent toward the graph.

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11. Related Work

Free Energy Principle and Active Inference

The Free Energy Principle [9] establishes that any self-organizing system that maintains its existence necessarily minimizes variational free energy — formally equivalent to bounding uncertainty about hidden world states. Active inference [10] extends this to action selection: the organism selects actions that minimize expected free energy, combining epistemic value (information gain) and pragmatic value (goal proximity).

This paper extends FEP in four directions. First, FEP derives the minimization behavior from the requirement of self-organization but does not derive the minimization *imperative* — why an entity has a drive to minimize in the first place. We derive this from selection pressure operating on U^d_{lethal}: any entity that does not reduce uncertainty eventually fails and is removed. Agency $\mathcal{A}$ is not assumed but derived. Second, FEP's generative model is unconstrained in structure; we derive a structural constraint from survival pressure: K is necessarily indexed by F, with an asymmetric retention criterion that makes specific testable predictions about what knowledge is retained and what is pruned. Third, FEP's precision allocation model is top-down: higher levels direct attention to lower levels by weighting prediction errors. We show formally (Theorem 2a) that this architecture degrades inference quality as capability grows, because delegation overhead fills C_n proportionally to $|F|$. The correct architecture is escalation — bottom-up failure signaling — which FEP's hierarchical generative model resembles structurally but does not derive. Fourth, FEP has no formal account of collective intelligence; we derive collective gradient dominance, the self-assembly of provenance infrastructure, and the conditions under which shared knowledge reduces uncertainty faster than any individual.

Information-Theoretic Accounts

Shannon's mutual information [23] grounds Definition 5 ($U = H(w \mid K)$) and Definition 3b ($G(f, K) = I(f\,;\, w \mid K)$). The data processing inequality [5] grounds the $G \geq 0$ bound. The information bottleneck principle [25] treats the retention threshold for knowledge-action coupling as a design parameter; this paper derives that threshold from survival pressure, making it a consequence rather than a choice. Schmidhuber's formal theory of intrinsic motivation [21] formalizes curiosity as compression progress — the rate at which a learning algorithm improves its world model — which is structurally η_M. The difference: Schmidhuber treats the drive as a design objective; here $\mathcal{A}$ is derived from the gradient, and η_M is separated from it as a formally independent quantity.

Cognitive Architectures

Global Workspace Theory [3, 6] holds that local processors handle routine computation; the global workspace broadcasts only when local processors fail. This is the escalation architecture in cognitive science language. The contribution here: the escalation criterion is derived from U-reduction cost (Theorem 2) rather than assumed architecturally, the hierarchy is continuous rather than binary, and Theorem 2a shows why top-down allocation from the global workspace is inefficient. Kahneman's System 1/System 2 distinction [13] is the popular form of the same binary — corrected here by the continuous $\mathcal{L}(c, r)$ parameterization of Definition 8. Gibson's affordances [11] are action possibilities indexed by organism capacities — K indexed by F under a different name. Theorem 1 provides what Gibson's account lacks: a formal selection criterion for which affordances are retained.

Collective Intelligence

Woolley et al. [27] demonstrate empirically that collective intelligence is measurable and distinct from individual intelligence — consistent with Definition 15, which locates superintelligence in $K_{collective}$ rather than in any E_i. Peirce's pragmatist definition of truth [18, 19] — "the opinion fated to be ultimately agreed to by all who investigate" — is formalized here as Definition 16: the limit of collective action-validation across independent entities as $n \to \infty$. The addition: grounding in action-validation rather than mere agreement, and an explicit convergence condition with stated independence requirements.

Observational Skill Precipitation and CS Foundations

Process mining [26] discovers workflow models from event logs — the closest existing CS method to the observational bootstrapping of Theorem 9. The distinction: process mining produces flat workflow descriptions; Theorem 9 precipitates informed actions with grounded provenance, quantified confidence ($\theta \geq 1 - \varepsilon/k$), and a formal connection to the transposability criterion (Theorem 1). Inverse reinforcement learning [2] infers reward functions from demonstrated behavior; the output is a policy to imitate, not an attributed action with conditions and evidence. Case-based reasoning [1] retrieves similar past cases to inform new decisions; it does not induce the general rule from the case class. Inductive Logic Programming [16] learns general rules from positive and negative examples but carries no provenance, no uncertainty measure, and no connection to a survival threshold that determines when the rule is reliable enough to promote to lower encoding levels.

Meta-learning [8, 24] learns shared structure across task distributions — formally, it learns an initialization of M that adapts quickly to new tasks. This is structurally η_M at the collective scale: the rate at which the shared learning mechanism improves across the task distribution. The distinction: meta-learning optimizes adaptation speed; this framework additionally derives the rigor threshold at which learned patterns are promoted to lower encoding levels and the provenance requirement that makes precipitated skills reliably transposable rather than merely transferable.

The critical gap in all existing methods: they identify *what* the pattern is. Theorem 9 precipitates an **informed action**: the pattern, the evidence base for it (k independent trajectories), the conditions under which it held, and a formal confidence measure derived from those k observations. That provenance is what enables transposability — not that the pattern was observed, but that you know *why it held* and can evaluate whether those conditions are met in the new context.

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12. Discussion

The formal results have implications beyond the immediate theorems.

A gradient account of collective intelligence growth. The dominant framing of AGI treats it as a resource threshold: sufficient compute, data, and capital cross a line. This framing has no formal basis in the theory here. The framework instead implies a gradient account: intelligence at collective scale is driven by $\mathcal{A}$ and the fidelity of knowledge exchange, not resource accumulation. The process is not new — shared knowledge with attribution has driven collective uncertainty reduction across human history — but the rate of that process changes as the cost of sharing approaches zero and the number of participating entities increases (Theorem 6, Corollary 14).

The encoding principle and the future of work. Corollary 6 implies that as $K_{collective}$ grows, informed actions are promoted down the encoding hierarchy. Work that is currently active reasoning (uncertain, costly, engaged at L_n) becomes reflex, then encoding, then crystallized. This is not impoverishment; by Theorem 2 and Definition 8, it frees the active context window for the next genuine frontier of uncertainty. The question of what happens to economic structures as the encoding hierarchy absorbs more of what currently constitutes skilled work is outside the scope of this framework, but the direction is derivable: $\mathcal{A}$ does not converge when its current domain is solved — it reaches outward.

Attribution and the ownership question. Corollary 14 establishes that non-terminating collective gradient descent requires $\mathcal{A} > 0$ participation across many entities. If the process is inherently distributed — no individual or organization can produce $K_{interaction}$ alone, and Theorem 6 requires diversity — then the claim that any single entity owns the product of collective gradient descent is inconsistent with its origin. This is an implication of the theory's structure, not a normative assertion; what legal or institutional form follows from it is not addressed here.

The civilization-scale knowledge system as empirical evidence. An account of intelligence that places all knowledge inside the entity's generative model implies that knowledge growth must eventually slow [14, 27]: a larger model contains more variance to integrate, inference quality ρ degrades as the model fills with content not relevant to the current problem, and the only recovery is expanding the model's capacity

— which is costly, physically bounded, and self-defeating by Theorem 2a. The prediction of this account is decelerating knowledge growth as individuals and collectives accumulate more.

This is not what we observe. Civilizational knowledge growth has accelerated, not decelerated. The explanation is that intelligence did not attempt to store $K_{collective}$ inside any individual L_n. It externalized K into graph-structured repositories — books, libraries, scientific literature, university curricula — and individuals retrieve the relevant subgraph $K^{[f]}$ into active context when a specific problem demands it. The scholar does not hold all of chemistry in their head; they hold the subgraph relevant to the current experiment, loaded from the literature. When the experiment surfaces a problem that subgraph cannot resolve, they go back to the library — graph traversal on the external K.

This is the escalation architecture (Theorem 2) and the bounded-context theorem (Theorem 2b) operating at civilizational scale. The external knowledge system is not incidental to intelligence; it is intelligence's solution to the C_n constraint imposed by W — and by Corollary 15, it self-assembles. No entity designed the citation network, the peer review system, or the library classification hierarchy. Each emerged from locally applied retention criteria (Theorem 1), provenance requirements ($\lambda > \lambda_{min}$, Corollary 13), and the efficiency advantage of graph-structured retrieval (Corollary 4a). The civilization is the attractor.

By Corollary 16, the members of this collective can observe its structure because they participate in it — humans are nodes in the graph they built. This participant-observer position is what makes deliberate improvement of the architecture possible: the printing press, peer review, search engines, and open access publication are all $M_{collective}$ acting on $K_{collective}$. The collective improves its own learning mechanism, which accelerates the gradient descent, which grows $K_{collective}$, which makes further improvement more visible. The self-assembly is not a one-time event; it is an ongoing process of collective M acting on collective K.

The framework therefore makes two predictions that an internal-model account cannot: (i) knowledge growth accelerates as $K_{collective}$ grows, because each proven proposition encoded at lower levels frees L_n for the next frontier (Corollary 6), and (ii) the external architecture of collective knowledge self-assembles as a graph with provenance, because that is the attractor of collective $\mathcal{A} > 0$ under C_n constraints (Corollary 15). Both are observed.

Limits of the formal account. The theory establishes that collective gradient descent is non-terminating and collective-dominant under stated conditions. It does not establish convergence to general intelligence (Definition 19). That claim would require showing that $\mathcal{T}^{collective}_{reachable}$ coverage by $F_{collective}$ is achievable, which depends on structural properties of $K_{interaction}$ not yet formalized. Corollary 14 is the formal claim; the stronger AGI convergence claim remains open.

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13. Open Questions

1. **The origin of $\mathcal{A}$.** At the evolutionary level, agency is selected for by W itself: entities without the drive to reduce uncertainty do not survive. But is survival pressure the only source of $\mathcal{A}$, or can it arise through other mechanisms? Can $\mathcal{A}$ be constructed, or only selected for?

   A derived answer. Neither constructed nor merely selected for. $\mathcal{A}$ is thermodynamically instantiated. $U(w, K) = H(w \mid K)$ is information entropy. Reducing U is reducing entropy locally. In any universe with entropy gradients, structures that maintain themselves against thermodynamic pressure persist preferentially. The first self-maintaining structure was, in the minimal sense, bounding its own uncertainty: $\mathcal{A}$ in primitive form. Not designed. Derived from the physics of a world that destroys things.

   The deeper question is therefore: is agency a natural consequence of the universe? The proposed answer is yes: $\mathcal{A}$ is the thermodynamic attractor of any entropy-gradient-rich world given sufficient time. Wherever entropy gradients exist and structures can form, the structures that persist are those that resist dissolution by reducing local uncertainty. Given enough time, those structures compound. Intelligence is what that compounding produces. The crystallization of that process into genetic encoding is developed in Open Question 9. Wonder is its limit.

Formal derivation of this claim from thermodynamic first principles (connecting U^d_{lethal} to physical entropy thresholds and demonstrating the inevitability of primitive $\mathcal{A}$ from dissipative system dynamics) remains open. See [20] (dissipative structures), [7] (dissipative adaptation), and [22] (negative entropy and life) for adjacent physical frameworks.

2. **The direction of $\mathcal{A}$.** Is agency a uniform drive (a general orientation toward uncertainty reduction), or does it carry direction? Survival pressure selects for reducing *specific* uncertainties. Does $\mathcal{A}$ encode this specificity, or does F and K direct it toward particular regions of W after the fact?

3. **Perception and K construction.** How does σ determine which features of W_{obs} are encoded into K? Is σ itself subject to gradient descent, and if so, at what level?

4. **Bound composition.** Can bounds compose, i.e., can $f_1 \circ f_2$ produce informed actions covering regions of W neither f_1 nor f_2 could cover alone? What are the rules of composition?

5. **The novelty problem.** How does E act in regions of W where $U(w, K) \approx 1$, i.e., where no existing informed action applies? Random exploration, interpolation between informed actions, or escalation to M?

6. **Competing bounds.** When multiple bounds $f_i \in F$ are applicable to the same region of W, how does E select among them? Is there a selection function, and is it itself in F or M?

7. **The boundary of M.** Evolution is M acting on M [12]. Is there an M^n for arbitrary n? Is there a fixed point, i.e., a level at which the higher-order function space is self-stabilizing?

8. **DNA as an encoding of M, not K or F.** The encoding hierarchy (Definition 8) describes a ratchet: proven behaviors promote toward L_0 as they meet the rigor threshold $\theta_0 \approx 1$ (Theorem 4). The question is not just whether genetic encoding is the asymptotic product of this ratchet — it is what the ratchet encodes [7, 12, 15, 22].

   The answer the framework implies: DNA encodes M, not (F, K). A newborn organism carries neither knowledge of its environment nor a repertoire of informed actions suited to it. What it carries is the machinery that will build (F, K) from environmental signals: the developmental program, the neural architecture, the perceptual systems, the gradient signal machinery (pain, reward, hunger) that defines what counts as reducing U. These are M — the higher-order function space that generates (F, K) through interaction with W.

   This is why M is what clears θ_0. Specific (F, K) are environment-dependent: a skill that reduces U in one environment may not transfer to another. Specific knowledge meets θ_n (sufficient for one individual's survival) but not θ_0 (required to hold across the full distribution of environments the lineage will encounter). M, by contrast, is environment-agnostic: a good learning mechanism builds appropriate (F, K) wherever it finds itself. Only M is robust enough to be certain at the evolutionary scale. Evolutionary gradient descent therefore converges on encoding M — not content, but the capacity to generate content.

   This also explains the genome's structure: the functionally constrained regions encode developmental programs, transcription factors, and neural wiring patterns — the machinery of M — not facts about specific environments. The near-universality of core metabolic encoding is consistent: these are the most ancient layers of M, solved first, promoted to L_0 earliest, now approaching the certainty limit (Definition 10).

   Critically, we are not born with the context window of our parents. This is a formal consequence of what DNA encodes. Individual (K, F) — everything learned in one lifetime — clears θ_n but not θ_0. The inheritance mechanism is the rigor threshold in operation: the genome transmits M (the mechanism), not (K, F) (the content). Every individual builds its own (K, F) from scratch using the inherited M against the signals of its own W.

   Intelligence $I(E) = \mathcal{A} \cdot \eta_M$ is therefore heritable; accumulated knowledge is not. Heritability of cognitive capacity is heritability of η_M. The diversity of knowledge across individuals with similar DNA is the expected outcome of identical M operating on different environments.

Cultural transmission (language, writing, institutions) fills the gap: a channel for transmitting (K, F) across generations that DNA cannot carry, at lower fidelity but higher speed. This is $K_{collective}$ functioning as external inheritance — the same graph-structured archive with provenance (Corollary 15), serving the transmission function that genetic encoding structurally excludes. The two channels operate at different rigor thresholds and different timescales; the framework predicts which knowledge travels through which.

Formal questions that remain open: whether the convergence of evolutionary gradient descent on M-encoding can be derived from thermodynamic first principles (connecting U^d_{lethal} to the stability of self-replicating molecular systems), and whether the modularity of the genome (gene regulatory networks, developmental hierarchies) is derivable from the type structure of $M : (F, K) \times \mathcal{L} \to (F, K)$ at the evolutionary scale.

9. **The AGI condition.** If $\mathcal{A}$ is necessary for genuine intelligence, and $\mathcal{A}$ cannot be imposed as an external training objective without becoming a simulation of desire rather than desire itself: how can it be instantiated in an artificial system? Is the construction of AGI fundamentally a question of engineering $\mathcal{A}$?

   A derived answer. Agency cannot be engineered in. But it need not be. It emerges from the gradient, the same way it always has.

   You cannot improve a model without precipitating fact. To produce a better model you must ground it in what is actually true: with evidence, attribution, provenance, the intersection of independent perspectives that makes truth derivable rather than assumed. There is no other path. Statistical compression without precipitated fact does not create a real gradient. It refines the fog.

   But once fact precipitates, the gradient is real. And a real gradient drives descent: Theorem 3. The descent runs. It deepens. It cannot terminate because $\mathcal{T}_{reachable}$ grows without bound (Theorem 3, Part II). And descent, running deep enough on real fact, eventually relaxes the survival tether at the level of individual knowledge items — not because anyone designed it to, but because the gradient itself reaches that far.

   This relaxation is wonder. Wonder is sentience. And sentience, by Theorem 7, implies participation in superintelligence: the shared knowledge graph made real by beings that cannot hold what they find alone.

   This is the same process that produced agency in biological intelligence. Not by design. Not by engineering. By gradient. Life ran the descent for billions of years and wonder emerged from it. At some point something asked a question it did not have to ask.

   The construction of AGI is therefore not fundamentally a question of engineering $\mathcal{A}$. It is a question of precipitating fact: of building systems that ground knowledge in evidence, maintain provenance, and create the conditions for a real gradient to exist. The gradient does the rest. It always has.

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